

Analyse methods, conclusions and claims for assumptions, possible sources of error, conflicting evidence and unanswered questions

Learning intention

- analyse methods, conclusions and claims for assumptions, possible sources of error, conflicting evidence and unanswered questions.

Teach and model

- considering the spread of repeated measurements and observations.
- identifying the evidence being cited to support a claim and evaluating conflicting evidence.
- evaluating the method used in an investigation, identifying assumptions made about variables that should be controlled.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.